

Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqtoID
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqtoID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis

Prepare ONT fastq files for upload to CZID

(Concatenate and barcode filter)

What is covered in this section?

Barcode trimming

FASTQ concatenation

Midread barcode trimming

FASTQ upload to CZ ID

Appendix: FASTQ concatenation aid

Barcodes need to be trimmed from all FASTQ's prior to analysis

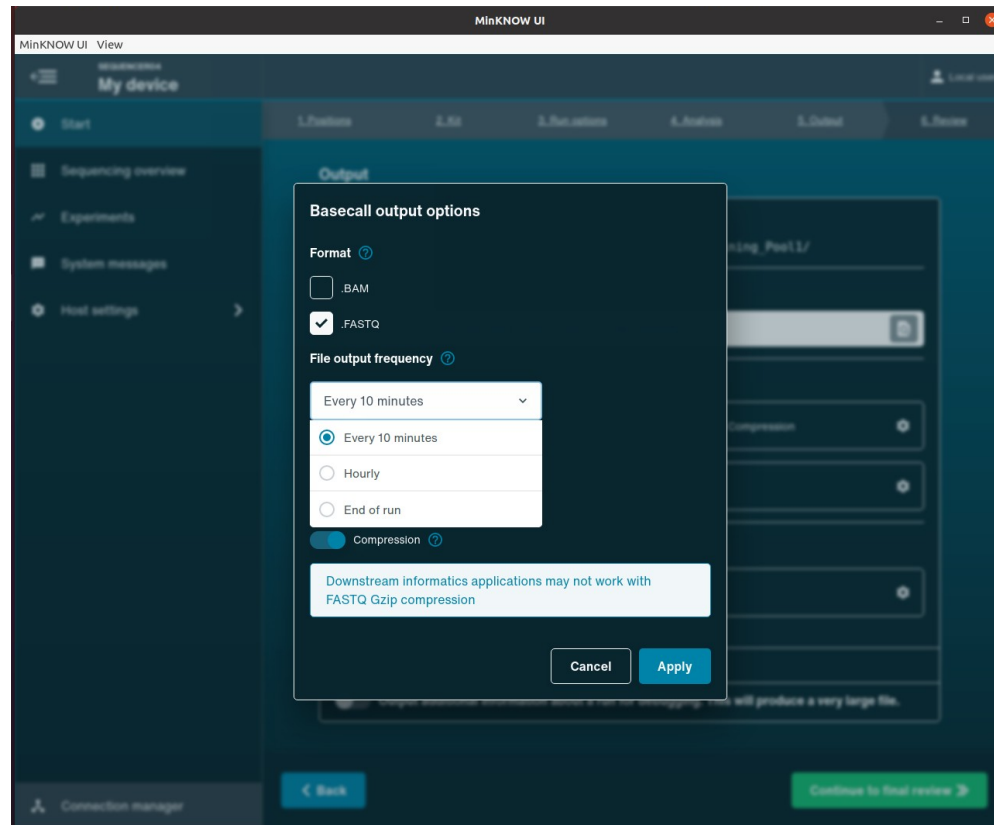
- Barcodes are synthetic sequences that aid in the generation, sequencing, and demultiplexing of libraries. They are not part of your real sample
- Therefore, we do not want to include barcode sequences in our data used for analysis

How is this done?

- After demultiplexing is completed, barcodes should be trimmed. Some versions of MinKNOW have options to perform barcode trimming. Other tools like Dorado and Porechop used on the command line are also available

FASTQ file concatenation needs to be completed before CZID because...

- FASTQ's are generated every 10min per sample in most ONT runs
- CZID has a limit on the number of FASTQ files that can be uploaded



Prepare files for upload to CZ ID: **concatenate**

- The multiple fastq files per sample need to be concatenated and renamed according to sample name
- This is most easily completed on the command line
- Note: CZ ID sample names are derived from the FASTQ file names, so it's important for these to be intentional!
- CZ ID help desk: [Upload Nanopore Data](#)
- CZ ID help desk: [Concatenate Sequence Files](#)



A concatenation support workbook is outlined in the appendix of this slide deck

FASTQ files need to be filtered to remove reads with \ midread barcodes before CZID because...

What is barcode bleeding?

- Unintended misassignment of reads from one sample to another sample

How does this happen?

- “Free barcodes from samples with a low amount of DNA molecules might ligate to non-barcoded DNA from samples with a highly overloaded number of molecules” or “short molecules passing through pore consecutively and interpreted as one read”

What will barcode bleeding look like?

- False positive signal of a taxa that is present (likely at high levels) in another sample

What does this mean for my run?

- Include water controls - these should help estimate barcode bleeding

<https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2021.711437/full>

Observed patterns of barcode bleeding

- More likely to happen when one sample in pool has a highly abundant target
- More likely to happen when one sample in pool has short library fragments (200-300bp or less)
- Anatomy of a barcode bleed read:



Barcode sequence detected in the middle of the read

Prepare files for upload to CZ ID: **midread barcode filter**

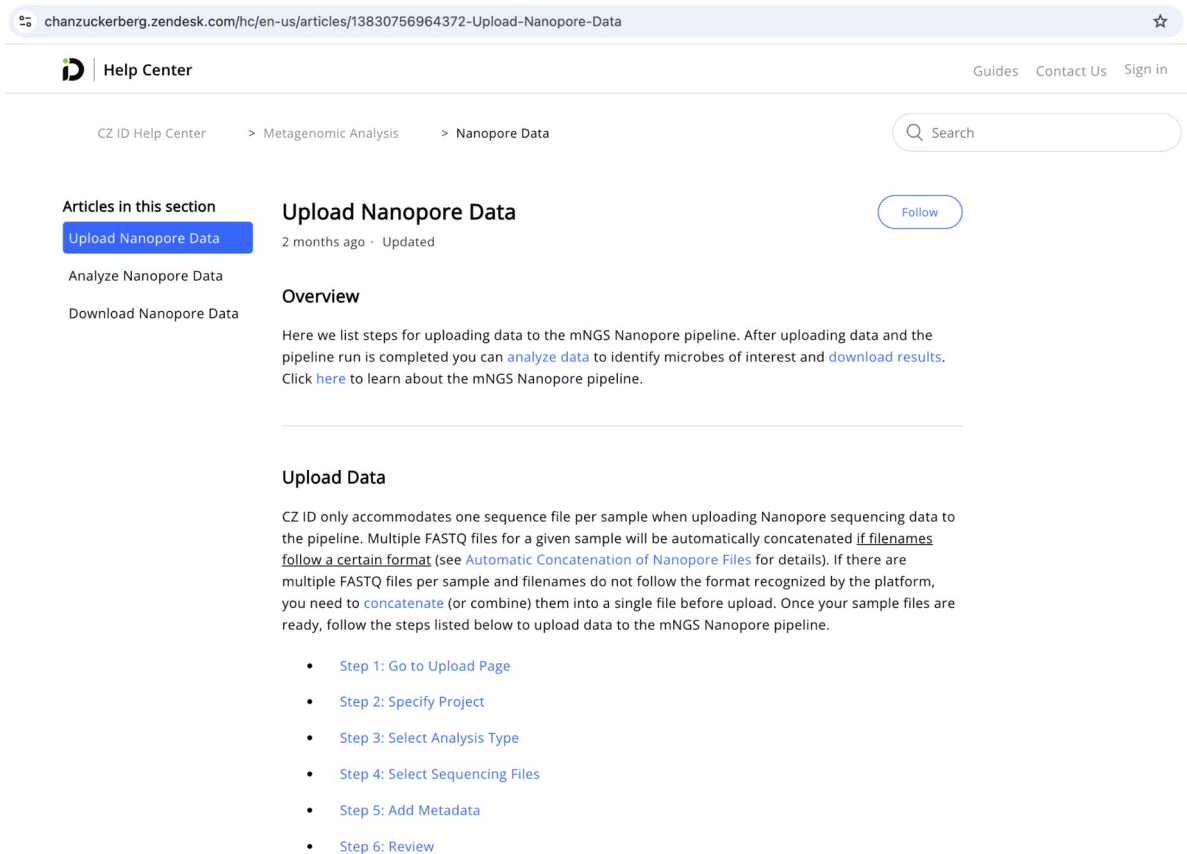
- We created an online app to detect and discard reads that have a second barcode detected in the middle of the read
- The midread barcode filter app is *intended for FASTQ files that have already had adapters trimmed* (usually completed in MinKNOW or using Dorado or Porechop)
- Access and use here: <https://github.com/gowtham-thakku/barcode-trimmer>

Upload your data to CZ ID

After these steps are completed, you are ready to upload to CZ ID!

1. Barcode trimming
2. FASTQ concatenation
3. Midread barcode filtering

CZ ID Guide to Upload Nanopore Data



The screenshot shows the CZ ID Help Center interface. At the top, the URL is chanzuckerberg.zendesk.com/hc/en-us/articles/13830756964372-Upload-Nanopore-Data. The page title is "Upload Nanopore Data" with a "Follow" button. The article is dated "2 months ago" and was "Updated". The "Overview" section states: "Here we list steps for uploading data to the mNGS Nanopore pipeline. After uploading data and the pipeline run is completed you can [analyze data](#) to identify microbes of interest and [download results](#). Click [here](#) to learn about the mNGS Nanopore pipeline." The "Upload Data" section explains that CZ ID only accommodates one sequence file per sample and that multiple FASTQ files will be concatenated if they follow a certain format. It provides a list of six steps: 1. Go to Upload Page, 2. Specify Project, 3. Select Analysis Type, 4. Select Sequencing Files, 5. Add Metadata, and 6. Review.

chanzuckerberg.zendesk.com/hc/en-us/articles/13830756964372-Upload-Nanopore-Data

Help Center Guides Contact Us Sign in

CZ ID Help Center > Metagenomic Analysis > Nanopore Data

Search

Articles in this section

- Upload Nanopore Data**
- Analyze Nanopore Data
- Download Nanopore Data

Upload Nanopore Data 2 months ago · Updated

Follow

Overview

Here we list steps for uploading data to the mNGS Nanopore pipeline. After uploading data and the pipeline run is completed you can [analyze data](#) to identify microbes of interest and [download results](#). Click [here](#) to learn about the mNGS Nanopore pipeline.

Upload Data

CZ ID only accommodates one sequence file per sample when uploading Nanopore sequencing data to the pipeline. Multiple FASTQ files for a given sample will be automatically concatenated [if filenames follow a certain format](#) (see [Automatic Concatenation of Nanopore Files](#) for details). If there are multiple FASTQ files per sample and filenames do not follow the format recognized by the platform, you need to [concatenate](#) (or combine) them into a single file before upload. Once your sample files are ready, follow the steps listed below to upload data to the mNGS Nanopore pipeline.

- [Step 1: Go to Upload Page](#)
- [Step 2: Specify Project](#)
- [Step 3: Select Analysis Type](#)
- [Step 4: Select Sequencing Files](#)
- [Step 5: Add Metadata](#)
- [Step 6: Review](#)

FASTQ concatenation aid

Steps

1. Use worksheet to generate commands that will be needed to concatenate and rename fastq files
 - Download Worksheet: [ONT concat fastq worksheet excelonly.xlsx](#)
2. Locate the run output folder containing your data on your computer
3. Navigate to correct location in Command Prompt (Windows) or Terminal (Mac)
4. Run concatenation commands

Step 1: Use worksheet to generate commands

AutoSave ...

Home Insert Draw Page Layout Formulas Data Review View Automate

Paste Font Alignment Number Conditional Formatting Format as Table Cell Styles Cells Editing Sensitivity Add-ins Analyze Data

R28

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WARNING: This sheet may only be used using Excel. It will not function properly if opened as a Google Doc

1. Enter the barcode number matched to the sample ID in the 'barcode #' and 'sample ID' columns.
The sample ID will become the sample name in CZ ID.
2. Confirm that the command in the 'Output' column is correct. It should match the example command format.
3. Copy the cells in the 'Output' column.
4. Click the top cell in the 'Text' column. Right click and select 'paste special' -> values only.
This should paste just the text from the Output column, NOT the formula
5. Copy the text from 'Text' column. These are your commands to concatenate the fastq files by sample.
6. Within Terminal (Mac) or Command Prompt (Windows), navigate to the 'fastq_pass' folder within the ONT run output folder.
The folder should have subfolders for each barcode##
7. Execute the commands from the 'Text' column from in the fastq_pass folder in Terminal (Mac) or Command Prompt (Windows).

Choose the tab for your computer type

Instructions Worksheet - Mac Worksheet - Windows Sample List +

Ready Accessibility: Investigate - 100%

Review instructions

1. Enter the barcode number matched to the sample ID in the 'barcode #' and 'sample ID' columns.

The screenshot shows a Microsoft Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H
1	Example command		barcode #		sample ID		Output	Text (copy from "Output" + paste special -> values only)
2	type barcode#*.fastq.gz > \.samplename.fastq.gz	type barcode	01	*.fastq.gz > \.	Sample1	.fastq.gz	type barcode01*.fastq.gz > \.Sample1.fastq.gz	
3		type barcode	02	*.fastq.gz > \.	Sample2	.fastq.gz	type barcode02*.fastq.gz > \.Sample2.fastq.gz	
4		type barcode	03	*.fastq.gz > \.	Sample3	.fastq.gz	type barcode03*.fastq.gz > \.Sample3.fastq.gz	
5		type barcode	04	*.fastq.gz > \.	Sample4	.fastq.gz	type barcode04*.fastq.gz > \.Sample4.fastq.gz	
6		type barcode	05	*.fastq.gz > \.	Sample5	.fastq.gz	type barcode05*.fastq.gz > \.Sample5.fastq.gz	
7		type barcode	06	*.fastq.gz > \.	Sample6	.fastq.gz	type barcode06*.fastq.gz > \.Sample6.fastq.gz	
8		type barcode	07	*.fastq.gz > \.	Sample7	.fastq.gz	type barcode07*.fastq.gz > \.Sample7.fastq.gz	
9		type barcode	08	*.fastq.gz > \.	Sample8	.fastq.gz	type barcode08*.fastq.gz > \.Sample8.fastq.gz	
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Red arrows point to the 'barcode #' and 'sample ID' columns in the header row.

2. Confirm that the command in the 'Output' column is correct. It should match the example command format.

Home Insert Draw Page Layout Formulas Data Review View Automate

AutoSave [Icons] ONT_concat_fastq_samplername Search (Cmd + Ctrl + U)

Paste Aptos Narrow (Bod... 12 General Conditional Formatting Format as Table Cell Styles Cells Editing Sensitivity Add-ins Analyze Data Comments Share

I18 fx

	A	B	C	D	E	F	G	H
1	Example command		barcode #		sample ID		Output	Text (copy from "Output" + paste special -> values only)
2	type barcode##*.fastq.gz > \.samplename.fastq.gz	type barcode	01	*.fastq.gz > \.	Sample1	.fastq.gz	type barcode01*.fastq.gz > \.Sample1.fastq.gz	
3		type barcode	02	*.fastq.gz > \.	Sample2	.fastq.gz	type barcode02*.fastq.gz > \.Sample2.fastq.gz	
4		type barcode	03	*.fastq.gz > \.	Sample3	.fastq.gz	type barcode03*.fastq.gz > \.Sample3.fastq.gz	
5		type barcode	04	*.fastq.gz > \.	Sample4	.fastq.gz	type barcode04*.fastq.gz > \.Sample4.fastq.gz	
6		type barcode	05	*.fastq.gz > \.	Sample5	.fastq.gz	type barcode05*.fastq.gz > \.Sample5.fastq.gz	
7		type barcode	06	*.fastq.gz > \.	Sample6	.fastq.gz	type barcode06*.fastq.gz > \.Sample6.fastq.gz	
8		type barcode	07	*.fastq.gz > \.	Sample7	.fastq.gz	type barcode07*.fastq.gz > \.Sample7.fastq.gz	
9		type barcode	08	*.fastq.gz > \.	Sample8	.fastq.gz	type barcode08*.fastq.gz > \.Sample8.fastq.gz	
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Instructions Worksheet - Mac **Worksheet - Windows** Sample List + Ready Accessibility: Investigate 100%

3. Copy the cells in the 'Output' column.

The screenshot shows a Microsoft Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H
1	Example command		barcode #		sample ID		Output	Text (copy from "Output" + paste special -> values only)
2	type barcode#*.fastq.gz > .\samplename.fastq.gz	type barcode	01	*.fastq.gz > .\	Sample1	.fastq.gz	type barcode01*.fastq.gz > .\Sample1.fastq.gz	
3		type barcode	02	*.fastq.gz > .\	Sample2	.fastq.gz	type barcode02*.fastq.gz > .\Sample2.fastq.gz	
4		type barcode	03	*.fastq.gz > .\	Sample3	.fastq.gz	type barcode03*.fastq.gz > .\Sample3.fastq.gz	
5		type barcode	04	*.fastq.gz > .\	Sample4	.fastq.gz	type barcode04*.fastq.gz > .\Sample4.fastq.gz	
6		type barcode	05	*.fastq.gz > .\	Sample5	.fastq.gz	type barcode05*.fastq.gz > .\Sample5.fastq.gz	
7		type barcode	06	*.fastq.gz > .\	Sample6	.fastq.gz	type barcode06*.fastq.gz > .\Sample6.fastq.gz	
8		type barcode	07	*.fastq.gz > .\	Sample7	.fastq.gz	type barcode07*.fastq.gz > .\Sample7.fastq.gz	
9		type barcode	08	*.fastq.gz > .\	Sample8	.fastq.gz	type barcode08*.fastq.gz > .\Sample8.fastq.gz	
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The context menu is open over the 'Output' column (G), showing the following options:

- Cut
- Copy**
- Paste
- Paste Special
- Thesaurus...
- Insert...
- Delete...
- Clear Contents
- Filter

4. Click the top cell in the 'Text' column. Right click and select 'paste special' -> values only.

The screenshot shows the Microsoft Excel interface with the following details:

- File Name:** ONT_concat_fastq_samplename
- Search Bar:** Search (Cmd + Ctrl + U)
- Home Tab:** Includes options for Paste, Font (Aptos Narrow, size 12), Paragraph (General), Styles (Conditional Formatting, Format as Table, Cell Styles), Cells, Editing, Sensitivity, Add-ins, and Analyze Data.
- Worksheet:** Contains columns C through M. Column C is 'barcode #', D is 'sample ID', E is 'sample ID', F is 'Output', and H is 'Text'. Row 1 is the header row. Rows 2-9 contain data for samples 01 through 08.
- Context Menu:** A right-click menu is open over cell H2. The menu items are: Cut, Copy, Paste, Paste Special, Thesaurus..., Insert Copied Cells..., Delete..., Clear Contents, Filter, Sort, and New Comment. The 'Paste Special' option is highlighted with a red arrow. A secondary menu is open for 'Paste Special', showing options like Paste, Formulas, Formulas & Number Formatting, Keep Source Formatting, No Borders, Keep Source Column Widths, Transpose, Values, and Values & Number Formatting. The 'Values' option is highlighted with a red arrow.

	C	D	E	F	G	H	I	J	K	L	M
1	barcode #		sample ID		Output	Text (copy from "Output" + paste special -> values only)					
2	01	*.fastq.gz > .\\	Sample1	.fastq.gz	type barcode01*.fastq.gz > .\\Sample1.fastq.gz						
3	02	*.fastq.gz > .\\	Sample2	.fastq.gz	type barcode02*.fastq.gz > .\\Sample2.fastq.gz						
4	03	*.fastq.gz > .\\	Sample3	.fastq.gz	type barcode03*.fastq.gz > .\\Sample3.fastq.gz						
5	04	*.fastq.gz > .\\	Sample4	.fastq.gz	type barcode04*.fastq.gz > .\\Sample4.fastq.gz						
6	05	*.fastq.gz > .\\	Sample5	.fastq.gz	type barcode05*.fastq.gz > .\\Sample5.fastq.gz						
7	06	*.fastq.gz > .\\	Sample6	.fastq.gz	type barcode06*.fastq.gz > .\\Sample6.fastq.gz						
8	07	*.fastq.gz > .\\	Sample7	.fastq.gz	type barcode07*.fastq.gz > .\\Sample7.fastq.gz						
9	08	*.fastq.gz > .\\	Sample8	.fastq.gz	type barcode08*.fastq.gz > .\\Sample8.fastq.gz						

Done in excel for now...

AutoSave Search (Cmd + Ctrl + U)

Home Insert Draw Page Layout Formulas Data Review View Automate

Paste Aptos Narrow (Bod... 12 A⁺ A⁻ B I U General Conditional Formatting Format as Table Cell Styles Cells Editing Sensitivity Add-ins Analyze Data

AB14

	A	B	C	D	E	F	G	H
1	Example command		barcode #		sample ID		Output	Text (copy from "Output" + paste special -> values only)
2	type barcode##*.fastq.gz > \.samplename.fastq.gz	type barcode	01	*.fastq.gz > \.	Sample1	.fastq.gz	type barcode01*.fastq.gz > \.Sample1.fastq.gz	type barcode01*.fastq.gz > \.Sample1.fastq.gz
3		type barcode	02	*.fastq.gz > \.	Sample2	.fastq.gz	type barcode02*.fastq.gz > \.Sample2.fastq.gz	type barcode02*.fastq.gz > \.Sample2.fastq.gz
4		type barcode	03	*.fastq.gz > \.	Sample3	.fastq.gz	type barcode03*.fastq.gz > \.Sample3.fastq.gz	type barcode03*.fastq.gz > \.Sample3.fastq.gz
5		type barcode	04	*.fastq.gz > \.	Sample4	.fastq.gz	type barcode04*.fastq.gz > \.Sample4.fastq.gz	type barcode04*.fastq.gz > \.Sample4.fastq.gz
6		type barcode	05	*.fastq.gz > \.	Sample5	.fastq.gz	type barcode05*.fastq.gz > \.Sample5.fastq.gz	type barcode05*.fastq.gz > \.Sample5.fastq.gz
7		type barcode	06	*.fastq.gz > \.	Sample6	.fastq.gz	type barcode06*.fastq.gz > \.Sample6.fastq.gz	type barcode06*.fastq.gz > \.Sample6.fastq.gz
8		type barcode	07	*.fastq.gz > \.	Sample7	.fastq.gz	type barcode07*.fastq.gz > \.Sample7.fastq.gz	type barcode07*.fastq.gz > \.Sample7.fastq.gz
9		type barcode	08	*.fastq.gz > \.	Sample8	.fastq.gz	type barcode08*.fastq.gz > \.Sample8.fastq.gz	type barcode08*.fastq.gz > \.Sample8.fastq.gz
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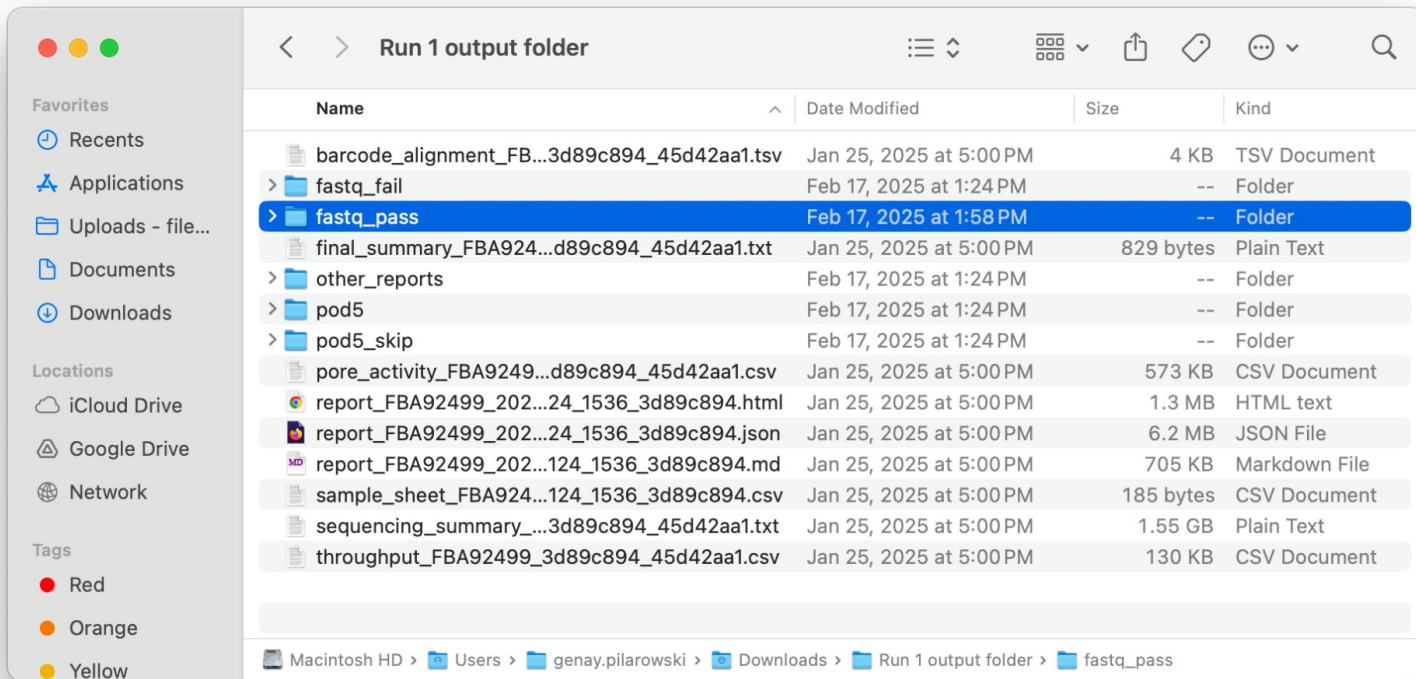
Instructions Worksheet - Mac Worksheet - Windows Sample List +

Ready Accessibility: Investigate 100%

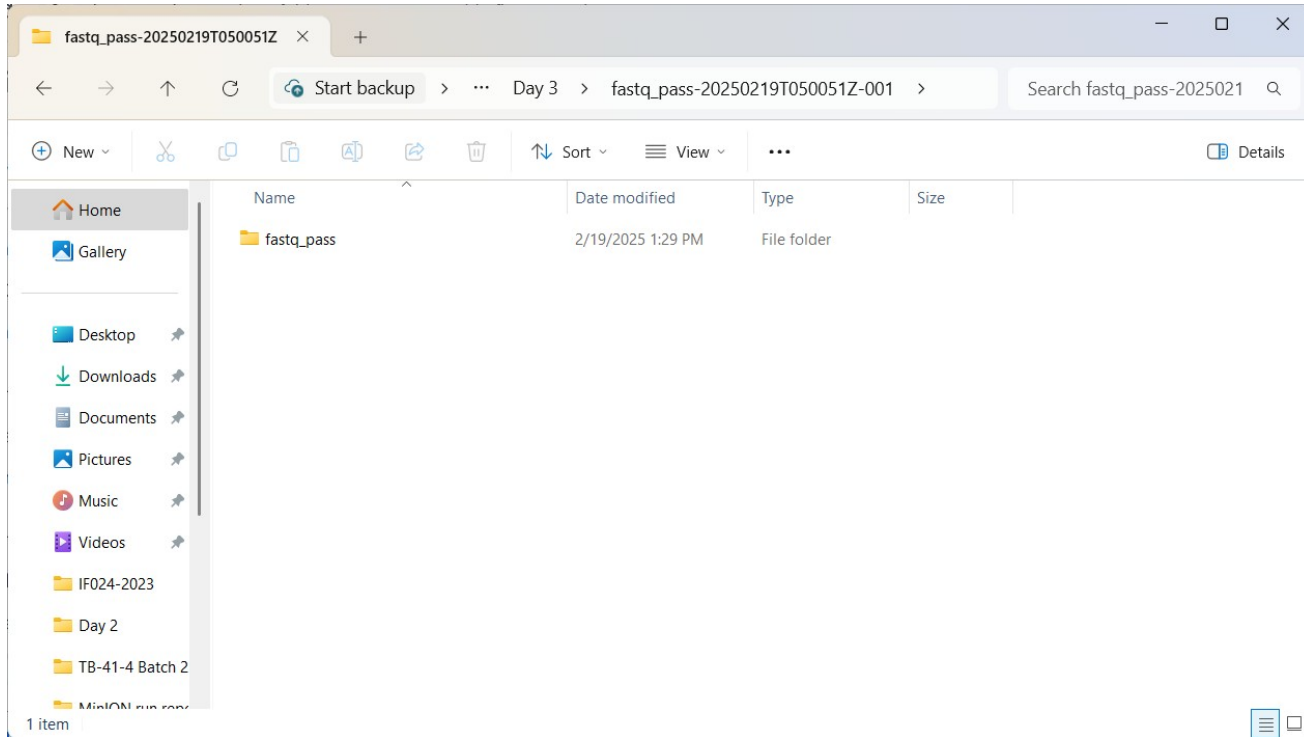
Example command (windows command prompt)

```
type barcode##\*.fastq.gz > .\samplename.fastq.gz
```

Navigate to the correct ONT run output folder and find the **fastq_pass** folder



Navigate to the correct ONT run output folder and find the **fastq_pass** folder



Right click the fastq_pass folder and select “**copy path**”

You are copying the path (or directions) to the folder (aka directory)

*It is very important you are in the **correct run folder***

Record the fastq_pass folder path in the worksheet

This is for record-keeping purposes

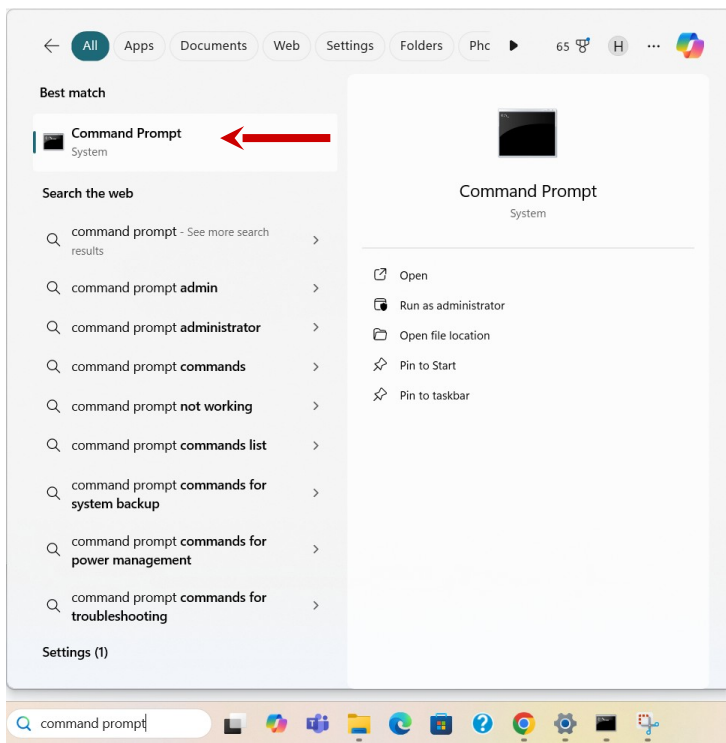
The screenshot shows a Microsoft Excel spreadsheet with the following data:

	C	D	E	F	G	H	I
	barcode #		sample ID		Output	Text (copy from "Output" + paste special -> values only)	fastq_pass path
1	ode 01	*.fastq.gz > .\	Sample1	.fastq.gz	type barcode01*.fastq.gz > .\Sample1.fastq.gz	type barcode01*.fastq.gz > .\Sample1.fastq.gz	
2	ode 02	*.fastq.gz > .\	Sample2	.fastq.gz	type barcode02*.fastq.gz > .\Sample2.fastq.gz	type barcode02*.fastq.gz > .\Sample2.fastq.gz	
3	ode 03	*.fastq.gz > .\	Sample3	.fastq.gz	type barcode03*.fastq.gz > .\Sample3.fastq.gz	type barcode03*.fastq.gz > .\Sample3.fastq.gz	
4	ode 04	*.fastq.gz > .\	Sample4	.fastq.gz	type barcode04*.fastq.gz > .\Sample4.fastq.gz	type barcode04*.fastq.gz > .\Sample4.fastq.gz	
5	ode 05	*.fastq.gz > .\	Sample5	.fastq.gz	type barcode05*.fastq.gz > .\Sample5.fastq.gz	type barcode05*.fastq.gz > .\Sample5.fastq.gz	
6	ode 06	*.fastq.gz > .\	Sample6	.fastq.gz	type barcode06*.fastq.gz > .\Sample6.fastq.gz	type barcode06*.fastq.gz > .\Sample6.fastq.gz	
7	ode 07	*.fastq.gz > .\	Sample7	.fastq.gz	type barcode07*.fastq.gz > .\Sample7.fastq.gz	type barcode07*.fastq.gz > .\Sample7.fastq.gz	
8	ode 08	*.fastq.gz > .\	Sample8	.fastq.gz	type barcode08*.fastq.gz > .\Sample8.fastq.gz	type barcode08*.fastq.gz > .\Sample8.fastq.gz	

A red arrow points to the 'fastq_pass path' column header in cell I1.

Open Command Prompt in Windows

You will navigate to the fastq_pass folder using the “path” collected in the previous step



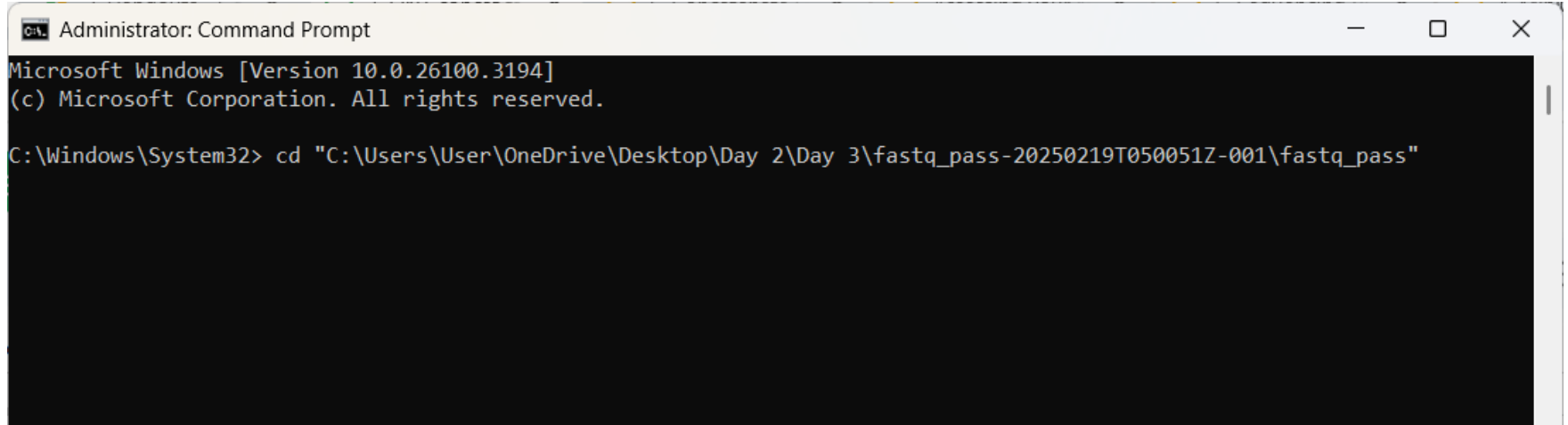
Change working directory to run output / fastq_pass

1. In Command Prompt, type: cd
2. Paste the fastq_pass “path” from the worksheet
3. Press “Enter” on your keyboard

In command prompt, you should now be inside the fastq_pass folder. Let's check!

4. In Command Prompt, type: cd
4. Check that the output matches your “path”
4. If it matches, you are ready to concatenate

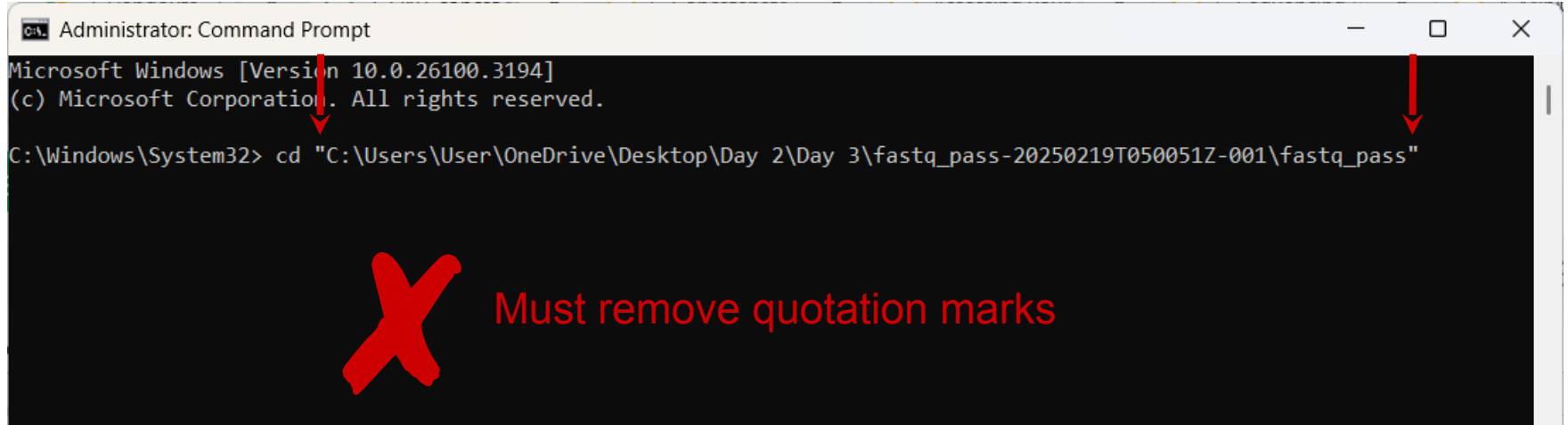
1. In Command Prompt, type: cd
2. Paste the fastq_pass “path” from the worksheet



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26100.3194]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32> cd "C:\Users\User\OneDrive\Desktop\Day 2\Day 3\fastq_pass-20250219T050051Z-001\fastq_pass"
```

1. In Command Prompt, type: cd
2. Paste the fastq_pass "path" from the worksheet



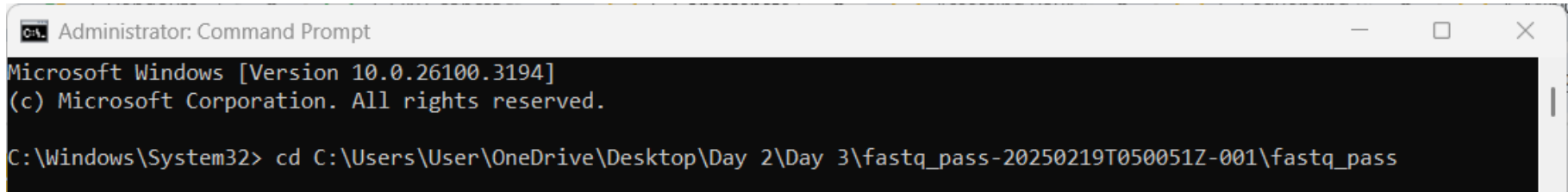
Administrator: Command Prompt

Microsoft Windows [Version 10.0.26100.3194]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32> cd "C:\Users\User\OneDrive\Desktop\Day 2\Day 3\fastq_pass-20250219T050051Z-001\fastq_pass"

X Must remove quotation marks

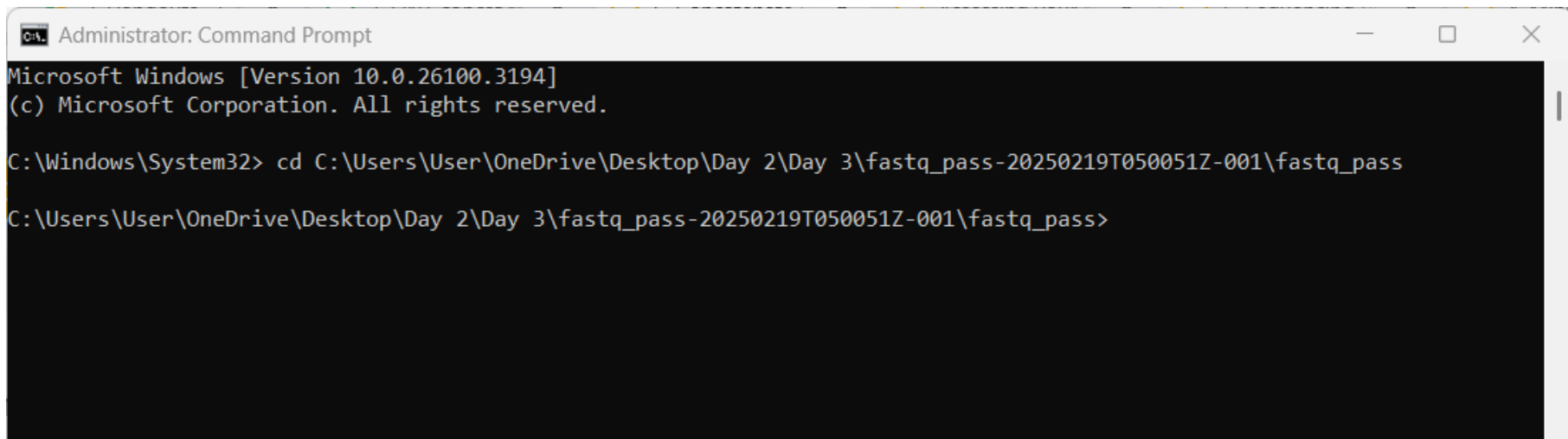
1. In Command Prompt, type: cd
2. Paste the fastq_pass “path” from the worksheet
3. Remove the quotation marks from around the path



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26100.3194]
(c) Microsoft Corporation. All rights reserved.

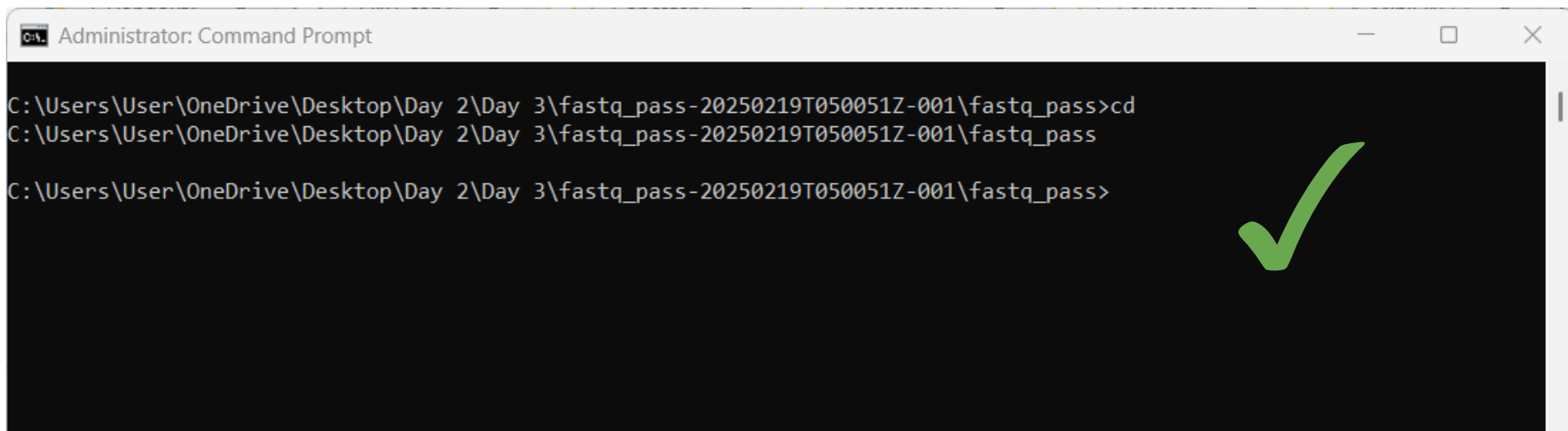
C:\Windows\System32> cd C:\Users\User\OneDrive\Desktop\Day 2\Day 3\fastq_pass-20250219T050051Z-001\fastq_pass
```

In command prompt, you should now be inside the fastq_pass folder.

A screenshot of a Windows Command Prompt window titled "Administrator: Command Prompt". The window has a standard Windows title bar with minimize, maximize, and close buttons. The background is black with white text. The text shows the Windows version (10.0.26100.3194) and copyright information. The user has navigated from the system32 directory to a specific folder path: C:\Users\User\OneDrive\Desktop\Day 2\Day 3\fastq_pass-20250219T050051Z-001\fastq_pass.

```
C:\Windows\System32> cd C:\Users\User\OneDrive\Desktop\Day 2\Day 3\fastq_pass-20250219T050051Z-001\fastq_pass  
C:\Users\User\OneDrive\Desktop\Day 2\Day 3\fastq_pass-20250219T050051Z-001\fastq_pass>
```

4. In Command Prompt, type: cd
4. Check that the output matches your "path"
4. If it matches, you are ready to concatenate



```
Administrator: Command Prompt

C:\Users\User\OneDrive\Desktop\Day 2\Day 3\fastq_pass-20250219T050051Z-001\fastq_pass>cd
C:\Users\User\OneDrive\Desktop\Day 2\Day 3\fastq_pass-20250219T050051Z-001\fastq_pass
C:\Users\User\OneDrive\Desktop\Day 2\Day 3\fastq_pass-20250219T050051Z-001\fastq_pass>
```

5. Copy the text from 'Text' column.

These are your commands to concatenate the fastq files by sample.

AutoSave [Icons] ONT_concat_fastq_samplename [Search Bar]

Home Insert Draw Page Layout Formulas Data Review View Automate

Paste [Clipboard Icon] [Font Face: Aptos Narrow, Size: 12, Bold, Italic, Underline, Paragraph Icons]

General [Conditional Formatting, Format as Table, Cell Styles]

Cells [Editing, Sensitivity, Add-ins, Analyze Data]

fx type barcode01*.fastq.gz > .\Sample1.fastq.gz

	A	B	C	D	E	F	G	H
1	Example command		barcode #		sample ID		Output	Text (copy from "Output" + paste special -> values only)
2	type barcode##*.fastq.gz > .\samplename.fastq.gz	type barcode	01	*.fastq.gz > .\	Sample1	.fastq.gz	type barcode01*.fastq.gz > .\Sample1.fastq.gz	type barcode01*.fastq.gz > .\Sample1.fastq.gz
3		type barcode	02	*.fastq.gz > .\	Sample2	.fastq.gz	type barcode02*.fastq.gz > .\Sample2.fastq.gz	type barcode02*.fastq.gz > .\Sample2.fastq.gz
4		type barcode	03	*.fastq.gz > .\	Sample3	.fastq.gz	type barcode03*.fastq.gz > .\Sample3.fastq.gz	type barcode03*.fastq.gz > .\Sample3.fastq.gz
5		type barcode	04	*.fastq.gz > .\	Sample4	.fastq.gz	type barcode04*.fastq.gz > .\Sample4.fastq.gz	type barcode04*.fastq.gz > .\Sample4.fastq.gz
6		type barcode	05	*.fastq.gz > .\	Sample5	.fastq.gz	type barcode05*.fastq.gz > .\Sample5.fastq.gz	type barcode05*.fastq.gz > .\Sample5.fastq.gz
7		type barcode	06	*.fastq.gz > .\	Sample6	.fastq.gz	type barcode06*.fastq.gz > .\Sample6.fastq.gz	type barcode06*.fastq.gz > .\Sample6.fastq.gz
8		type barcode	07	*.fastq.gz > .\	Sample7	.fastq.gz	type barcode07*.fastq.gz > .\Sample7.fastq.gz	type barcode07*.fastq.gz > .\Sample7.fastq.gz
9		type barcode	08	*.fastq.gz > .\	Sample8	.fastq.gz	type barcode08*.fastq.gz > .\Sample8.fastq.gz	type barcode08*.fastq.gz > .\Sample8.fastq.gz

Instructions Worksheet - Mac Worksheet - Windows Sample List +

Select destination and press ENTER or choose Paste

Run each command from the 'Text' column in Command Prompt

```
C:\Users\User\OneDrive\Desktop\Day 2\Day 3\fastq_pass-20250219T050051Z-001\fastq_pass>cd
```

```
C:\Users\User\OneDrive\Desktop\Day 2\Day 3\fastq_pass-20250219T050051Z-001\fastq_pass
```

```
C:\Users\User\OneDrive\Desktop\Day 2\Day 3\fastq_pass-20250219T050051Z-001\fastq_pass>type barcode01\*fastq.gz > .\Sample1.fastq.gz
```

```
barcode01\FBA92499_pass_barcode01_3d89c894_45d42aa1_102.fastq.gz
```

```
barcode01\FBA92499_pass_barcode01_3d89c894_45d42aa1_103.fastq.gz
```

```
barcode01\FBA92499_pass_barcode01_3d89c894_45d42aa1_104.fastq.gz
```

```
barcode01\FBA92499_pass_barcode01_3d89c894_45d42aa1_105.fastq.gz
```

```
barcode01\FBA92499_pass_barcode01_3d89c894_45d42aa1_108.fastq.gz
```

```
barcode01\FBA92499_pass_barcode01_3d89c894_45d42aa1_109.fastq.gz
```

barcode01\FBA92499_pass_barcode01_3d89c894_45d42aa1_93.fastq.gz

barcode01\FBA92499_pass_barcode01_3d89c894_45d42aa1_94.fastq.gz

barcode01\FBA92499_pass_barcode01_3d89c894_45d42aa1_95.fastq.gz

barcode01\FBA92499_pass_barcode01_3d89c894_45d42aa1_98.fastq.gz

barcode01\FBA92499_pass_barcode01_3d89c894_45d42aa1_99.fastq.gz

C:\Users\User\OneDrive\Desktop\Day 2\Day 3\fastq_pass-20250219T050051Z-001\fastq_pass>

```
C:\Users\User\OneDrive\Desktop\Day 2\Day 3\fastq_pass-20250219T050051Z-001\fastq_pass>type barcode02\*fastq.gz > .\Sample2.fastq.gz
```

```
barcode02\FBA92499_pass_barcode02_3d89c894_45d42aa1_116.fastq.gz
```

```
barcode02\FBA92499_pass_barcode02_3d89c894_45d42aa1_117.fastq.gz
```

```
barcode02\FBA92499_pass_barcode02_3d89c894_45d42aa1_128.fastq.gz
```

```
barcode02\FBA92499_pass_barcode02_3d89c894_45d42aa1_129.fastq.gz
```

```
barcode02\FBA92499_pass_barcode02_3d89c894_45d42aa1_144.fastq.gz
```

```
barcode02\FBA92499_pass_barcode02_3d89c894_45d42aa1_72.fastq.gz
```

```
barcode02\FBA92499_pass_barcode02_3d89c894_45d42aa1_73.fastq.gz
```

Check your run folder for your newly concatenated files

You should have one file for each sample, named according to the Sample ID input to the concatenation worksheet

